

Neural-Field Fusion Preserves a Buried Shared Factor that Pooling Discards

Background

Fusing time-series modalities is hard when the link between them is fine, low-energy, and buried in a multiscale background. Three strategies compete. *Late fusion* encodes each modality and pools to a vector — cheap, but pooling discards the fine, temporally-localized component carrying the link. *Early fusion* cross-attends over raw tokens — faithful but scales badly with length. *Neural-field (NF) fusion* fits a per-modality implicit neural representation (INR) to each signal and fuses on the field latents; a reconstruction objective should force the latent to retain the fine component, and fields queried in a shared continuous coordinate fuse naturally across mismatched sampling grids. In a falsification-first design we ask: does NF fusion preserve the buried shared factor better than pooling, and reach early-fusion accuracy at lower cost?

Method

A controlled generator emits two heterogeneous 1-D modalities — a frequency-modulated chirp (A) and an amplitude-modulated tone (B) — that share one slowly-varying instantaneous-frequency trajectory $f(t)$; a pair is labeled 1 iff A and B carry the same $f(t)$. The matching component is low-energy within a $1/f^\beta$ multiscale background (an SNR knob), with different sampling rates and jitter. The dataset is pre-validated: the label is joint-only (high-capacity single-modality classifiers sit at chance), per-modality marginals match across classes, and an oracle on $f(t)$ is near-perfect while a classical time-frequency baseline beats chance. Four families are compared at matched parameters, steps, optimizer, and data, with FLOPs logged separately for representation vs fusion: late (pooled), early (cross-attention), and two amortized neural fields — LAINR (per-modality locality-aware multi-band INR plus a small fusion head) and OmniField (cross-modal conditioned field). A linear probe regresses $f(t)$ from each frozen representation to isolate the mechanism. We sweep SNR over three seeds with shuffled-pair and unimodal controls.

Results

Four predictions were pre-registered. P1 — the NF advantage over late fusion widens with difficulty: SUPPORTED, under strict parameter matching. As SNR falls 12→−6 dB the accuracies coincide at the extremes and separate between; late collapses to chance by 3 dB while LAINR holds 0.76–0.87 through 0 and −3 dB (gap up to +0.35). Given $5.7\times$ more parameters, late *still* collapses — the advantage is the field's inductive bias, not capacity. P3 — $f(t)$ is more decodable from the NF latent: SUPPORTED. Even where accuracies match, linear-probe R^2 is ≈ 0.9 for NF vs ≈ 0.6 for pooling, and the probe gap tracks the accuracy collapse: the difference is representational, set at encoding. P2 — NF reaches early-fusion accuracy at lower fusion cost: SUPPORTED. LAINR matches or exceeds early accuracy at $\sim 1800\times$ fewer fusion FLOPs and, at matched parameters, the lowest total compute (Pareto-dominant). P4 — the per-modality LAINR beats the cross-modal OmniField across the regime. Controls remain at chance (no leakage). Boundaries are honest: at very high SNR all methods tie, and at −6 dB all fall to chance (below a classical baseline) — the advantage is bounded to an intermediate-difficulty band.

Conclusion

In low-SNR, nonstationary, mismatched-grid correspondence problems, reconstruction-trained per-modality neural-field latents preserve the fine shared factor that pooling discards — giving large, mechanistically-explained robustness at lower fusion cost, shown under strict parameter matching with controls. The effect is specific to the per-modality field design (not neural fields broadly) and to an intermediate-difficulty band, and the efficiency holds on the fusion axis (representation cost rises). Practically: when a shared latent is faint and modalities are sampled incompatibly, fuse on neural-field latents, not pooled embeddings. The harness is falsification-first and reproducible, and the same 1-D pipeline transfers to real heterogeneous signals such as ECG+PPG cardiac fusion.